

THE STATISTICAL MECHANICS OF SIEVE-THEORETIC PRIMES: THE CRITICAL SPECTRUM, THE LOG-POWER DICHOTOMY, AND TWO FAMILIES THAT ARE NEVER UNIQUE

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ABSTRACT. We compute the thermodynamics of the classical sieve-theoretic prime families for the localized Bost–Connes systems of [6]. The critical exponents proposed are all correct, the open/closed dichotomy has a clean and complete explanation, and the account of high-temperature uniqueness is wrong for two of the six families, for an elementary reason that had not been noticed.

Critical exponents. If $\pi_S(x) \asymp x^\theta / \log^k x$ then Abel summation gives $\beta_c(S) = \theta$, and at $\beta = \theta$ the series behaves like $\int^x dt / (t \log^k t)$. Hence $\beta_c = 1$ for twin, Sophie Germain and Chen primes; $3/4$ for Friedlander–Iwaniec $a^2 + b^4$; $2/3$ for Heath-Brown $a^3 + 2b^3$; $1/2$ for Landau $n^2 + 1$; and $1/c$ for Piatetski-Shapiro $[n^c]$, a continuous spectrum. All six values are as proposed.

The dichotomy is exactly the log-power. The critical series converges iff $k > 1$, so the low-temperature phase is closed precisely for the families defined by *two* simultaneous prime conditions — twin, Sophie Germain, Chen, where $k = 2$ — and open for those defined by a single condition on a polynomial or fractional sequence, where $k = 1$. This is a complete answer, and it identifies the source of closedness as the second prime condition rather than as anything about the sieve.

Two of the six are congruence-obstructed, and are never unique — and which two is decided by a kernel condition, not by confinement alone. All four congruence-trapped families sit in a single residue class: twin and Sophie Germain primes in $2 \bmod 3$, Landau and the odd Friedlander–Iwaniec primes in $1 \bmod 4$. But confinement obstructs only when the occupied class lies in the *kernel* of the character. For Landau and Friedlander–Iwaniec the class is 1, the kernel itself, so the nontrivial character mod 4 has *finite* detecting set and lies in Ξ_β for every β : a permanent obstruction in the sense of [6, §4.1], and **for these two families the KMS $_\beta$ state is never unique, at any temperature**. For twin and Sophie Germain primes the opposite happens: $\chi(2) = -1$ for the character mod 3, so the detecting set is *cofinite* in S and the character is detected at full strength — the class 2 generates $(\mathbb{Z}/3)^\times$, and by [6, Lem. 5.6] the admissible classes generate $(\mathbb{Z}/M\mathbb{Z})^\times$ for every modulus, so no congruence obstructs these families at all. Verified numerically to $2 \cdot 10^5$ (single classes as stated).

Twin, Sophie Germain, Chen, Heath-Brown and Piatetski-Shapiro primes are therefore unobstructed, and for them the equidistribution criterion is the right one wherever the critical series diverges.

The finding is consistent with [6, §5]: Theorem 5.5(5) there, with Lemma 5.6, already established that the Sophie Germain family is confined to no proper kernel, and the present analysis confirms it — the mod-3 confinement, far from obstructing, feeds the detecting series in full.

1. THE CRITICAL EXPONENT

The framework and notation are those of [6], cited by DOI; no unpublished companion paper is used.

Proposition 1.1 (Abel summation). *Let $S \subseteq P$ satisfy $\pi_S(x) \asymp x^\theta / \log^k x$ with $\theta \in (0, 1]$, $k \geq 0$. Then*

$$(1.1) \quad \sum_{p \in S, p \leq x} p^{-\beta} \asymp \int_2^x \frac{t^{\theta-1-\beta}}{\log^k t} dt,$$

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XXIV-sieve/](#).

so that $\beta_c(S) = \theta$, and at the critical value $\beta = \theta$ the right side is $\int_2^x dt/(t \log^k t)$, which converges as $x \rightarrow \infty$ if and only if $k > 1$.

Proof. The counting function has density $\asymp \theta t^{\theta-1}/\log^k t$; substitute this into the Stieltjes integral $\sum_{p \leq x} p^{-\beta} = \int_2^x t^{-\beta} d\pi_S(t)$. For $\beta < \theta$ the integral is $\asymp x^{\theta-\beta}/\log^k x \rightarrow \infty$; for $\beta > \theta$ it converges. At $\beta = \theta$, $\int dt/(t \log^k t) = [\log^{1-k} t/(1-k)]$ for $k \neq 1$ and $\log \log t$ for $k = 1$. $\square$

Theorem 1.2 (The spectrum).

S	θ	k	$\beta_c(S)$	phase
<i>twin</i> $\{p : p+2 \in P\}$	1	2	1	<i>closed</i> $[1, \infty)$
<i>Sophie Germain</i> $\{p : 2p+1 \in P\}$	1	2	1	<i>closed</i> $[1, \infty)$
<i>Chen</i> $\{p : p+2 \in P_2\}$	1	2	1	<i>closed</i> $[1, \infty)$
<i>Friedlander–Iwaniec</i> $a^2 + b^4$	3/4	1	3/4	<i>open</i> $(3/4, \infty)$
<i>Heath–Brown</i> $a^3 + 2b^3$	2/3	1	2/3	<i>open</i> $(2/3, \infty)$
<i>Landau</i> $n^2 + 1$	1/2	1	1/2	<i>open</i> $(1/2, \infty)$
<i>Piatetski–Shapiro</i> $[n^c]$	1/c	1	1/c	<i>open</i> $(1/c, \infty)$

Proof. Proposition 1.1 with the counting functions: Brun’s sieve [4] gives the upper bounds with $k = 2$ for the shifted families, and Chen’s theorem [1] the matching lower bound for Chen primes; the Friedlander–Iwaniec [2] and Heath–Brown [3] asymptotics are theorems with $k = 1$; the Landau asymptotic with $k = 1$ is conjectural (Bateman–Horn); the Piatetski–Shapiro asymptotic with $k = 1$ is a theorem for $c < 243/205$ by Rivat–Sargos [5]. $\square$

Remark 1.3 (The dichotomy is the second prime condition). By Proposition 1.1 the phase is closed exactly when $k > 1$, and $k = 2$ occurs exactly for the families imposing *two* simultaneous primality conditions — p and $p+2$, p and $2p+1$, p and $p+2 \in P_2$. The polynomial and fractional families impose one condition and have $k = 1$. So the closedness observed for Sophie Germain primes in [6, Thm. 5.5] is not a feature of the sieve but of the second condition, and it transfers verbatim to twins and to Chen primes, the latter unconditionally since Chen’s theorem supplies the lower bound that the twin case still lacks.

2. CONGRUENCE CONFINEMENT VERSUS OBSTRUCTION

Lemma 2.1 (Elementary congruences). (1) If $p > 3$ and $p+2$ is prime then $p \equiv 2 \pmod{3}$.
(2) If $p > 3$ and $2p+1$ is prime then $p \equiv 2 \pmod{3}$.
(3) If $p = n^2 + 1$ is prime and $p > 2$ then n is even and $p \equiv 1 \pmod{4}$.
(4) Squares and fourth powers are $\equiv 0, 1 \pmod{4}$, so $a^2 + b^4 \equiv 0, 1, 2 \pmod{4}$; hence an odd prime of the form $a^2 + b^4$ satisfies $p \equiv 1 \pmod{4}$.

Proof. (1) If $p \equiv 1$ then $3 \mid p+2$. (2) If $p \equiv 1$ then $3 \mid 2p+1$. (3) If n is odd then $n^2 + 1 \equiv 2 \pmod{4}$, forcing $p = 2$; so n is even and $p = 4m^2 + 1$. (4) Direct. $\square$

Theorem 2.2 (Permanent obstruction — and its exact scope). Let χ_4 be the nontrivial character modulo 4 and χ_3 the nontrivial character modulo 3; for a Dirichlet character, $|1 - \chi(\sigma_p)| = |1 - \chi(p)|$, the Frobenius convention of [6, (1.1)] inverting $\chi(p)$ without changing the modulus of the summand. Then:

- (1) For S the Landau primes $n^2 + 1$ or the Friedlander–Iwaniec primes $a^2 + b^4$, the detecting set $D_{\chi_4} = \{p \in S : \chi_4(p) \neq 1\}$ is finite, since by Lemma 2.1(3)–(4) all odd members lie in the class 1 mod 4, which is $\ker \chi_4$. Hence $\sum_{D_{\chi_4}} p^{-\beta} < \infty$ for every $\beta > 0$, $\chi_4 \in \Xi_\beta(\mathbb{Q}, S)$ for every β , the KMS_β simplex has at least two extreme points at every temperature, and the state is never unique.

- (2) For S the twin or Sophie Germain primes the same confinement has the opposite effect: by Lemma 2.1(1)–(2) all members $p > 3$ lie in the class $2 \bmod 3$, and $\chi_3(2) = -1 \neq 1$, so D_{χ_3} is cofinite in S and $\sum_{D_{\chi_3}} p^{-\beta} = \sum_{p \in S} p^{-\beta} + O(1)$: the character is detected at full strength. More generally the class 2 generates $(\mathbb{Z}/3\mathbb{Z})^\times$, and by [6, Lem. 5.6] the admissible classes of these families generate $(\mathbb{Z}/M\mathbb{Z})^\times$ for every modulus M , so no Dirichlet character has finite detecting set: there is no permanent congruence obstruction for twin or Sophie Germain primes.

Proof. A character trivial on the single occupied class has finite detecting set — that is (1), where the class is the kernel itself — and a character nontrivial on it detects all but finitely many members — that is the first half of (2). For the last claim of (2), a character with finite detecting set would be trivial on all but finitely many of S , hence on the subgroup generated by the occupied classes, which is everything by [6, Lem. 5.6]. The never-uniqueness in (1) is [6, Thm. 3.9], uniqueness requiring $\sum_{D_\chi} p^{-\beta} = \infty$ for every nontrivial χ ; the obstruction is permanent in the sense of [6, §4.1]. $\square$

Remark 2.3 (Numerical verification). Computed to $2 \cdot 10^5$, counting $p > 3$ for the mod-3 rows, odd p for Landau, all representable primes for Friedlander–Iwaniec, $p > 3$ with $p + 2 \in P_2$ for Chen, positive a, b for Heath-Brown (the range of [3]), and the distinct primes $[n^c] \leq 2 \cdot 10^5$ for Piatetski-Shapiro:

S	$\#S$	M	coprime classes occupied
twin (smaller member)	2159	3	$\{2\}$ of $\{1, 2\}$
Sophie Germain	2056	3	$\{2\}$ of $\{1, 2\}$
Landau $n^2 + 1$	63	4	$\{1\}$ of $\{1, 3\}$
Friedlander–Iwaniec	938	4	$\{1\}$ of $\{1, 3\}$
Chen	7523	3, 4	both classes
Heath-Brown	289	4, 9	both mod 4; $\{1, 2, 7, 8\} \bmod 9$
Piatetski-Shapiro $c = 1.1$	6040	3, 4	both classes

The single-class rows now illustrate *both* halves of Theorem 2.2: for Landau and Friedlander–Iwaniec the occupied class is the kernel $\{1\}$ (obstruction); for twin and Sophie Germain it is $\{2\}$, on which χ_3 is -1 (full-strength detection). For Heath-Brown modulo 9 the set $\{1, 2, 7, 8\}$ is not contained in a proper subgroup of $(\mathbb{Z}/9\mathbb{Z})^\times$: a character trivial on it must have $\chi(2) = 1$ and hence be trivial, since 2 generates. So Heath-Brown is not obstructed either.

Corollary 2.4 (The proposed criterion is not the right one). *The claim that high-temperature uniqueness is equivalent to equidistribution across admissible coprime classes is correct in one direction — equidistribution does force $\sum_{D_\chi} p^{-\beta} \asymp \zeta_S(\beta) = \infty$ for $\beta < \beta_c$, hence uniqueness — but it is not necessary: twin and Sophie Germain primes are as far from equidistributed as possible, confined to one class, and are nevertheless unobstructed, the confining class generating the group. What decides is not equidistribution but [6, Thm. 5.5(5)]: confinement to a proper kernel. That happens for Landau and Friedlander–Iwaniec, and for no other family here.*

Remark 2.5 (Consequence for [6]). [6, Thm. 5.5] establishes for the Sophie Germain family the Brun-summability, the closed low-temperature phase, and — in its part (5), with [6, Lem. 5.6] — the uniqueness criterion in the divergent range. All of it stands, and Theorem 2.2(2) is its confirmation from the congruence side: the mod-3 confinement feeds the detecting series rather than starving it. The correct picture for Sophie Germain primes is therefore: no permanent obstruction anywhere; uniqueness wherever the series diverges (conditional on the lower bound of [6, Thm. 5.5(2)]); and full symmetry breaking, $\Xi_\beta = \widehat{G}_S$, on the closed phase $[1, \infty)$ where ζ_S converges.

3. THE CONTINUOUS SPECTRUM

Theorem 3.1 (Piatetski-Shapiro). *For $1 < c < 243/205$ the asymptotic $\pi_{\text{PS}(c)}(x) \sim x^{1/c}/\log x$ is a theorem (Piatetski-Shapiro for $c < 12/11$, Heath-Brown for $c < 755/662$, Rivat–Sargos for $c < 243/205$). Hence for those c the system $\mathcal{A}_{\mathbb{Q}, \text{PS}(c)}$ has $\beta_c = 1/c \in (205/243, 1)$, unconditionally, with an open low-temperature phase; and $c \mapsto 1/c$ traces a continuous family of critical temperatures.*

Remark 3.2 (The range is narrow). $205/243 = 0.8436$, so the unconditional range of critical temperatures is $\beta_c \in (0.8436, 1)$, an interval of length about 0.156. The full conjectural range $(0, 1)$ requires the asymptotic for all $c > 1$, which is open. It is worth stating the unconditional interval numerically, since “a continuous spectrum on $(0, 1)$ ” overstates what is available.

4. ASSESSMENT

Claim	Status	Reference
$\beta_c = 1$ for twin, SG, Chen	true	Thm. 1.2
$\beta_c = 3/4, 2/3, 1/2$ for FI, HB, Landau	true	Thm. 1.2
$\beta_c = 1/c$ for Piatetski-Shapiro	true	Thm. 3.1
closed iff Brun-summable	true; and iff $k > 1$	Prop. 1.1
closedness comes from the sieve	no: from the 2nd prime condition	Rem. 1.3
uniqueness $\iff$ equidistribution	no: kernel confinement decides	Cor. 2.4
Landau, FI: unique somewhere	false: never unique	Thm. 2.2(1)
twin, SG obstructed	no: class 2 generates	Thm. 2.2(2)
Chen, HB, PS obstructed	no	Rem. 2.3
PS spectrum covers $(0, 1)$	unconditionally only $(0.8436, 1)$	Rem. 3.2

Two of the three proposed pieces are correct and one is not, and the failure is elementary rather than deep. The critical exponents are exactly the counting exponents; the open/closed dichotomy is exactly the power of the logarithm, hence exactly the number of simultaneous prime conditions. Four of the six families are trapped in a single residue class by a congruence that has nothing to do with sieves — but confinement obstructs only when the class sits in the kernel. That happens for Landau and Friedlander–Iwaniec, whose KMS states are never unique by [6, Thm. 3.9]; for twin and Sophie Germain primes the same confinement is detection at full strength, and there is no obstruction at all.

The moral is worth recording because it is the opposite of the usual one in this corpus. Here the analytic input — the hard sieve theorems — is not where the difficulty lies; those theorems give the critical exponents and the phase types cleanly. What decides the qualitative behaviour is a congruence available to a first-year student, read correctly: Ξ_β measures how a set of primes sits inside residue classes *relative to the kernels of characters*. A set confined to the class 1 is maximally obstructed no matter how delicate its density; a set confined to a generating class is not obstructed at all.

Two questions. For the closed-phase families — twin, Sophie Germain, Chen, all now known unobstructed — high-temperature uniqueness coexists with the closed phase as soon as the divergence of the critical series below β_c is available, which is unconditional for Chen (Chen’s theorem [1] supplies the lower bound) and conditional for the other two; can the Chen case be carried out in full, making it the first family in the corpus with both a closed phase and proven high-temperature uniqueness? And second, for the two obstructed families the permanent obstruction has order 2; is the full Ξ_β for, say, the Landau primes equal to that order-two group below β_c , or is it larger?

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